Nicholas Gowin- Github: NicholasGowin1 nigo6803@colorado.edu
Simon Logan- Github: SimonLogan323 silo1904@colorado.edu
Tariq Ahmad- Github: tariq-ahmad7 taah5903@colorado.edu
Justin Pleva- GitHub: jupl3767-spec jupl3767@colorado.edu
Akshay Pulaganti- GitHub: akpu1080 akpu1080@colorado.edu

Section: 013 - Group: 6

Team Name: Bloomers

Application Name: Bloom

Application Description:

Bloom is an interactive plant tracking app that gets people outside to find the wild plants of their own region. Each user gets a personalized checklist of plants native to or commonly found in their state, and they fill it in by photographing each one in the real world. Every confirmed find is saved to their account, building a personal field guide over time. This will use the PlantNet API. Users will take pictures of their local plants, and an AI will verify the plant species.

What sets us apart from other plant identification apps is that it gamifies the process. Not only does it teach people what plants are, but it also encourages this exploration. It makes going out and exploring fun. Users will be able to collect rewards for finding rare plants. Together, these features make Bloom both a fun challenge and a great way to connect with your environment.

Audience:

The designated audience for this application is nature lovers in Colorado who want to know the different plants that exist around them without having to look each one up on their own. This includes students, parents, teachers, hikers, and others who are curious about what they are seeing on trails, around campus, or in their own backyard. We anticipate that the age range for users will be anywhere from 10-75, so our UI will be very user-friendly to accommodate all ends of the spectrum.

Vision Statement:

To encourage users to learn about the nature surrounding them.

Repository Link:

https://github.com/SimonLogan323/Nature-ID-App

Version Control:

Done in GitHub.

Development Plan:

We will be using a hybrid set of practices drawing from several development methodologies, and we will be tracking our progress using this Group Kanban Board.

Communication Plan:

We will be combining in-person collaboration with communication through a group Discord server.

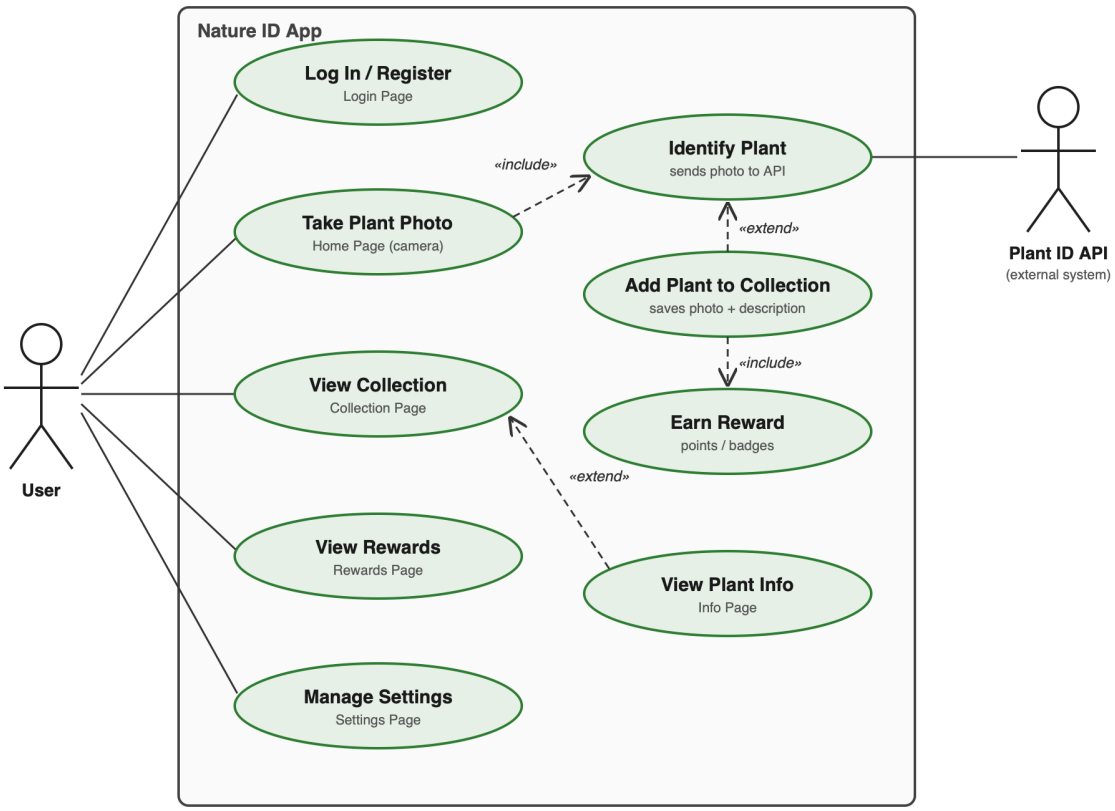
Meeting Plan:

Team Meeting: On Discord Tuesday afternoons.

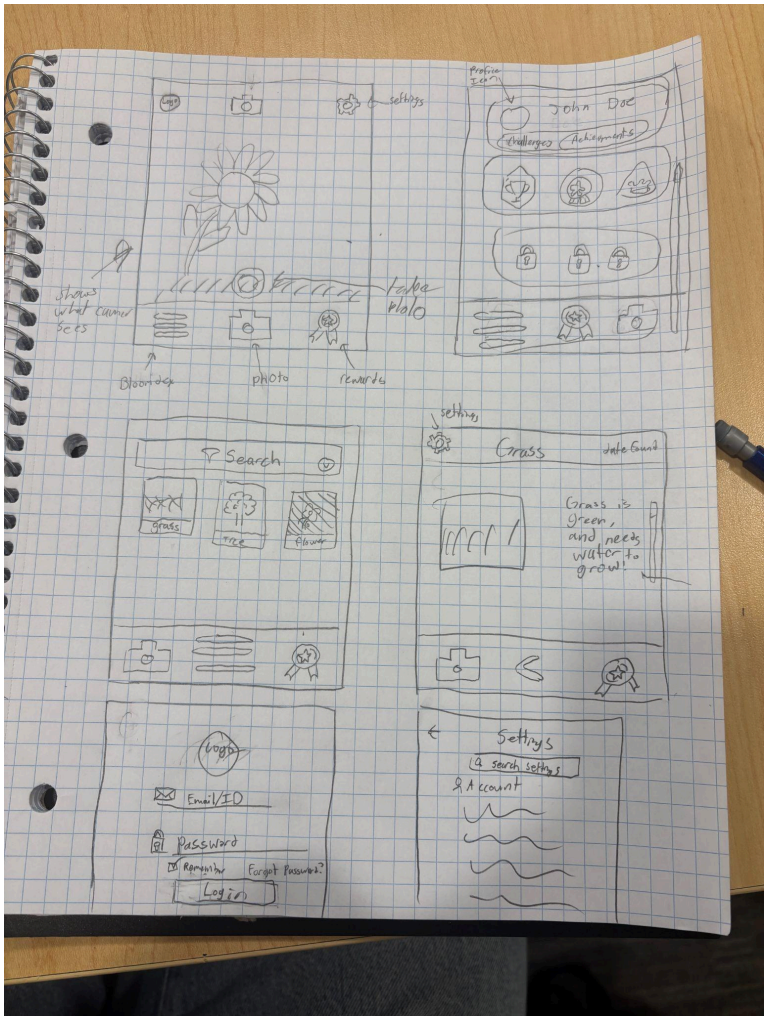
With TA: Monday at 2 pm over Zoom.

Case Diagram:

Use Case Diagram — Nature ID App



Wireframes:



EXTRA CREDIT:

Problem 1: If a user spams our page with images, it could disrupt our servers with tasks. This could be mitigated by deleting images if it is not a plant in the system. It will state that no plant was detected, and it will take them back to the home page, and if someone has a certain plant, their image will not be saved.

Problem 2: There could be code injection on the sign-in page. To solve this issue we would make it so that users would only be allowed to type a certain number of characters for each.

Problem 3: People might get too distracted by trying to find a plant/take a photo of a plant to the point where they are no longer aware of nature, private property, or other hazardous situations. To mitigate this issue we would utilize disclaimers/reminders in order to emphasize the importance of remaining aware for users.

Problem 4: While using the app users may get distracted and trample the plants, so we would need to add a disclaimer for users to respect nature in order to not harm or affect any ecosystems or habitats.

Problem 5: The API utilized by our application could hit rate limits which would break the apps intended purpose. To mitigate this issue we would make sure to add error handling such as a “try again later” message, in order to stop the API from hitting the free rate usage limits.